

Qianjun Xia's Curriculum Vitae

qx2253@columbia.edu | (917)295-7344 | qianjun-xia.github.io/ | www.linkedin.com/in/qianjunxia2002/

Education

- University of British Columbia**, PhD in Computer Science Sept 2026 – Present
- PhysAI Lab. Advisor: Prof. Peter Yichen Chen.
- Columbia University**, MS in Mechanical Engineering, Research Track Sept 2024 – May 2026
- Creative Machines Lab. Advisor: Prof. Hod Lipson.
 - GPA: 4.12/4.33
 - Concentration: Robotics and Control
- Shanghai Jiao Tong University**, BS in Mechanical Engineering and Mathematics Sept 2020 – Jun 2024
- Double Major: Mathematics and Applied Mathematics

Publications

- **DiagGen: Agentic Generation of Deformable Assets with Sim-based Diagnostics for Robotic Simulation.** G. Chen*, Y. Qu*, **Q. Xia***, P. Jing, Y. Cheng, B. Ma, P. Yang, B. Zhou, Z. Li, S. Suri, G. Sun, C. Liu, P. Y. Chen, Z. Zeng, F. Shi. *arXiv preprint*, 2026. [[arXiv](#)] [[Project Site](#)] (* equal contribution)
- **Agentic Real2Sim: Physics-based World Modeling with Vision-Language Agents.** G. Chen*, **Q. Xia***, J. Peng*, H. Zhang*, P. Jing*, B. Ma*, J. Qian, Y. Cheng, Z. Jiao, B. Zhou, Y. Qu, L. Ye, K. Zhang, K. Wang, W. Zeng, Y. Chen, P. Yang, Z. Zeng, S. Luo, H. Wang, C. Liu, A. Yuille, F. Shi, C. Zheng, Y. Li, C. Jiang, P. Y. Chen. *arXiv preprint*, 2026. [[arXiv](#)] [[Project Site](#)] [[Code](#)] (* equal contribution)
- **RoboVoxel: Learning Soft Body Dynamics via Video Colorization.** J. Lin*, **Q. Xia***, S. Zhang, H. Lipson. *Manuscript in preparation for journal submission*, 2026. (* equal contribution)
- **An MLP-Embedded Transformer-Physics Hybrid Framework for Inverse Design of Magnetic Rod.** H. Liu, **Q. Xia**, R. Wang, C. Huang, Z. Ullah, K. Xu, D. Wang. *Under review, IEEE Transactions on Automation Science and Engineering*, 2026.

Research Experience

- Agentic Real2Sim: Physics-based World Modeling with Vision-Language Agents** [[Project Page](#)] Mar 2026 – Sep 2026
University of British Columbia – PhysAI Lab. Advisor: Prof. Peter Yichen Chen
- Co-developed an agentic framework in which **vision-language agents** convert recordings of real robot-object interaction into **simulatable digital twins**, recovering scene geometry, object state, and physical parameters without per-scene hand tuning.
 - One unified Real2Sim pipeline covers rigid-object manipulation, **deformable-object interaction**, and **humanoid motion** – cases that previously required separate, specialized Real2Sim methods.
 - Resulting digital twins support downstream **policy learning** and policy evaluation in simulation; an **open-weight VLM** backend matches frontier-model success rates, making the pipeline cheap enough to run at dataset scale.
- DiagGen: Agentic Generation of Deformable Assets with Sim-based Diagnostics** [[Project Page](#)] Jun 2026 – Sep 2026
University of British Columbia – PhysAI Lab. Advisor: Prof. Peter Yichen Chen
- Improved the physical fidelity and stability of generated deformable assets in the **physics simulator** through systematic calibration of material and solver parameters and the introduction of safeguard bounds, yielding more plausible and robust object behavior under robotic manipulation.
 - Integrated DiagGen into the Agentic **Real2Sim** pipeline, enabling end-to-end generation of **simulation-ready scenes** directly from videos of a robot arm manipulating deformable objects.
- RoboVoxel: Soft-Body System Identification from Videos** [[Project Page](#)] Mar 2025 – Present
Columbia University – Creative Machines Lab. Advisors: Jiong Lin, Prof. Hod Lipson
- Built a physics-based **soft-body simulation** environment (elastic bodies under gravity and actuator forces) and used it to generate large-scale synthetic datasets spanning diverse geometries, material parameters, and actuation patterns to supervise video-based **system identification**.
 - Trained a **Video Transformer** that infers voxel-level soft-body physical parameters – stiffness, mass, and actuator layout – from multi-frame video, enabling video-driven digital-twin reconstruction that a simulator can directly replay.
 - Built a 4×4 physical voxel robot – 16 independently servo-driven active voxels with a gear-and-curved-rack mechanism – mirroring the simulator's actuation, and validated the model's **sim-to-real** system identification on real videos of the robot.

Shanghai Jiao Tong University. Advisor: Prof. Dong Wang. Manuscript in submission.

- Built a **Cosserat-rod** simulator that captures the full nonlinear dynamics of slender elastic structures – bending, twisting and shear – a general-purpose framework directly transferable to **cable, rope, and continuum-robot manipulation**.
- Extended the simulator with external magnetic body-torques to model a **magnetically actuated vascular guidewire**; Cosserat-rod predictions matched physical prototype behavior under a range of actuation profiles, enabling **model-based control** and sim-based policy design.
- Prototyped the hardware loop for **closed-loop validation**: a robot arm positions the driving magnet while a motor rotates the guidewire base; full control stack implemented in Python with SolidWorks-designed hardware and Xbox-controller teleoperation.

Projects

Rise of the AI Knight – World-Model RL for Action-Platformer Games [\[Project Page\]](#)

Sep 2025 – Present

Columbia University. Advisor: Prof. Shipra Agrawal

- Implemented a **world-model-based reinforcement-learning** agent for Hollow Knight and Hollow Knight: Silksong, adapting **latent-dynamics (Dreamer-style)** methods to a complex, real-time action-platformer environment with long-horizon credit assignment.

M.E.H: A Bipedal Robot [\[Project Page\]](#)

Mar 2025 – May 2025

Columbia University. Advisor: Prof. Hod Lipson

- Designed a parallel-linked legged robot and derived closed-form **inverse kinematics** for **gait planning**.
- Deployed a **real-time locomotion controller** on a Raspberry Pi, reaching a maximum walking speed of ~ 32 cm/s.

Industry Experience

Shanghai ABB Engineering Co., Ltd, Computer Vision Intern

Jun 2023 – Sept 2023

- Developed and tested **computer-vision** algorithms for industrial robotic automation alongside the perception team.
- Built a binocular-stereo vision demo in Halcon for automated tire-assembly positioning, achieving **1 mm 3D localization accuracy** – a real-to-sim-style calibration task bridging CAD models and camera observations.

Skills

Simulation & Physics: MuJoCo, Taichi, differentiable physics, soft-body dynamics, cable / deformable-object simulation

Robotics & ML: Video Transformers, VLA models, model-based reinforcement learning (world models), robot kinematics & dynamics

CAD & CAE: SolidWorks, Fusion360, UG, Catia, Adams, Ansys